Mapping Cells through Time and Space with moscot

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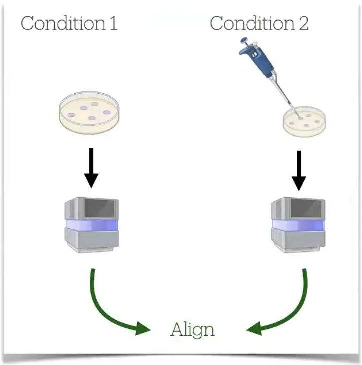
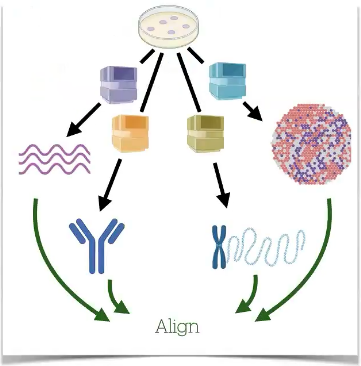
Why Cells have to be Realigned

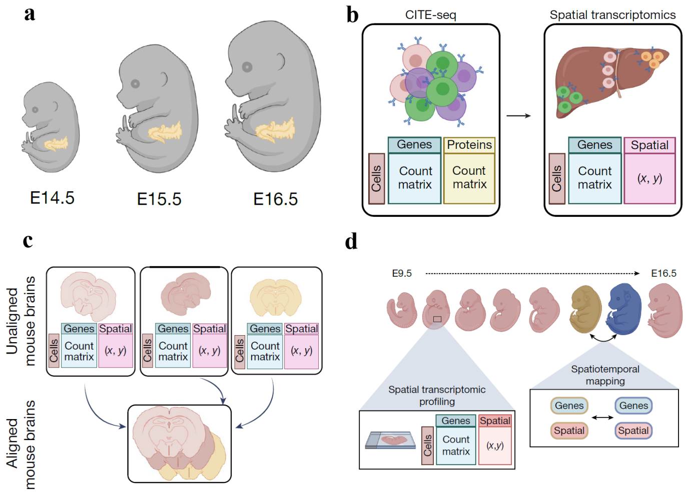
Sequencing technologies are destructive



Only a limited number of modalities can be measured

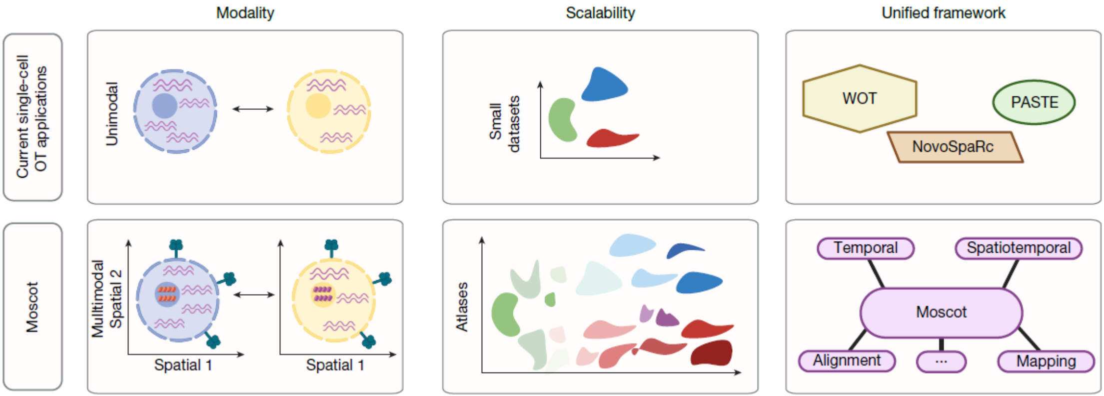


Alignment Tasks



- a) moscot.time
- b) moscot.space.mapping
- c) moscot.space.alignment
- d) moscot.spatialtemporal

Limitations of Existing Alignment Methods



moscot.time

Mathematical Formulation

$$\arg \min_{P \in \mathbb{R}_+^{N \times M}} \langle C, P \rangle + \frac{\epsilon \tau_a}{1 - \tau_a} \mathrm{KL}[P \mathbf{1}_M \parallel a] + \frac{\epsilon \tau_b}{1 - \tau_b} \mathrm{KL}\left[P^\top \mathbf{1}_N \parallel b\right] - \epsilon H(P)$$

- $H(P) = -\sum_{i,j} P_{ij}(log P_{ij} - 1)$
- N, M : Number of cells in t_1 and t_2
- $C \in \mathbb{R}^{N \times M}$: Cost matrix (based on molecular profiles)
- $P \in \mathbb{R}^{N \times M}$: Coupling matrix
- $a \in \mathbb{R}_+^{\mathbb{N}}, b \in \mathbb{R}_+^{\mathbb{M}}$: Prior distributions, $\sum_i a_i = \sum_j b_j = 1$
- ϵ, τ_a, τ_b : Hyperparameters

Adjusting the Marginals for Growth and Death

Define

$$a_i = \frac{g(x_i)^{t_2 - t_1}}{\sum_{j=1}^{N} g(x_j)^{t_2 - t_1}}, \quad \forall i \in \{1, \ldots, N\}$$

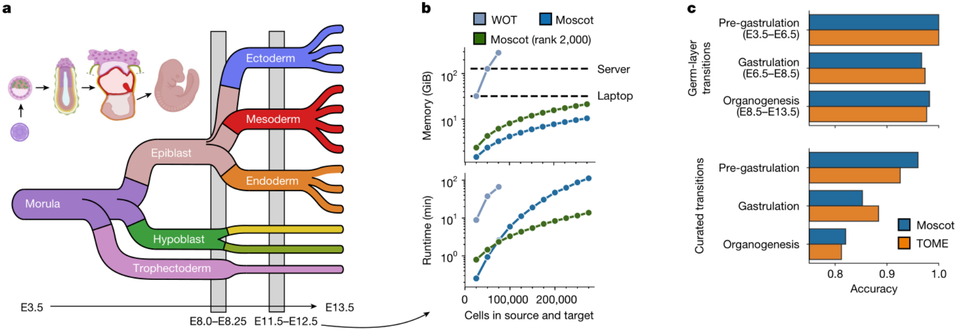
- $g : R^D \to R_+$: the expected value of a birth–death process
  $g(x) = exp\{\beta(x) - \delta(x)\}$
- Growth rate $\beta(x)$ and death rate $\delta(x)$ can be estimated using curated marker gene sets

moscot.time overvomes previous limitations

- (1) Multimodality: Given bimodal representations $(X^{(1)}, X^{(2)}), (Y^{(1)}, Y^{(2)})$, define the cost function C using both modalities:
  - a) Scale the modalities to have the same variance, and concatenate them to a joint feature space
  - b) Apply shared latent-space-learning techniques, such as VAEs

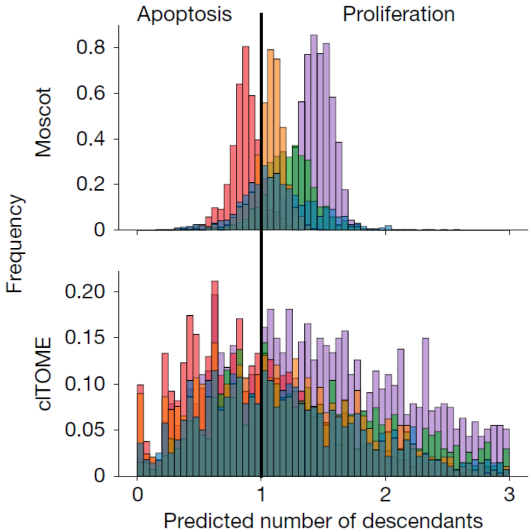
- (2) Scalability
  - a) Engineering improvements, such as GPU acceleration
  - b) Methodological innovation: low-rank approximation of coupling matrix P

Reconstructing Mouse Embryogenesis



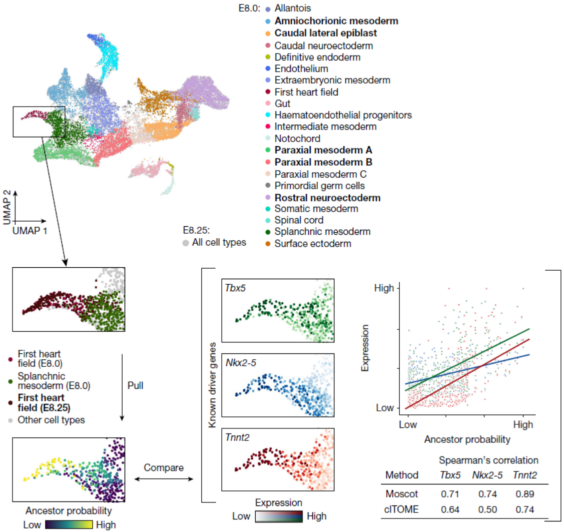
Published atlas of early mouse development that contains 1.7 million cells across 20 time points

Growth-rate Calculation



- Apoptosis rate estimation: $\hat{r} = \sum_{i=1}^{N}(1 - \frac{N}{M}\sum_{j=1}^{M} P_{ij})_{+}/N$
- Apply uniform marginals a and b
- Fix $\tau_b = 0.99995$, so that $\sum_i P_{ij} \approx 1$
- Tune τ_a such that the resulting apoptosis rates are biologically plausible
- clTOME predicts an apoptosis rate of 19%, which is biologically unrealistic
- Growth rate produced by moscot is more cell-type specific

Driver Gene Calculation



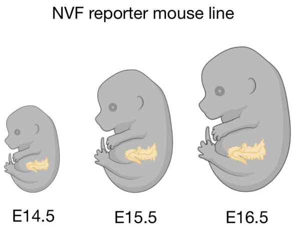
Method	Tbx5	Nkx2-5	Tnnt2
Moscot	0.71	0.74	0.89
cITOME	0.64	0.50	0.74

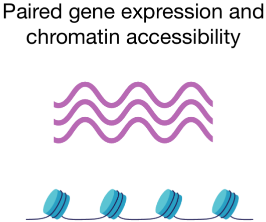
- For a cell type C, let c be the uniform distribution over cells in C. Then the pull-back distribution is defined as

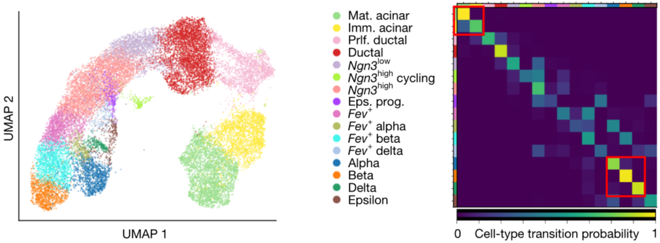
$$q = P^T c$$

- Correlate (Pearson / Spearman) gene expressions and the pull-back distribution q to obtain putative driver genes

Delineating Mouse Pancreas Development

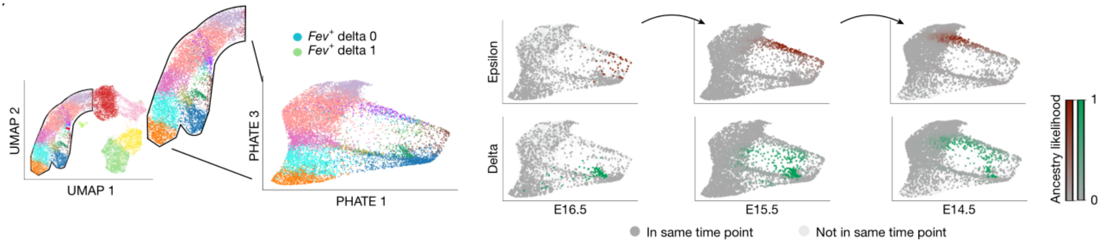






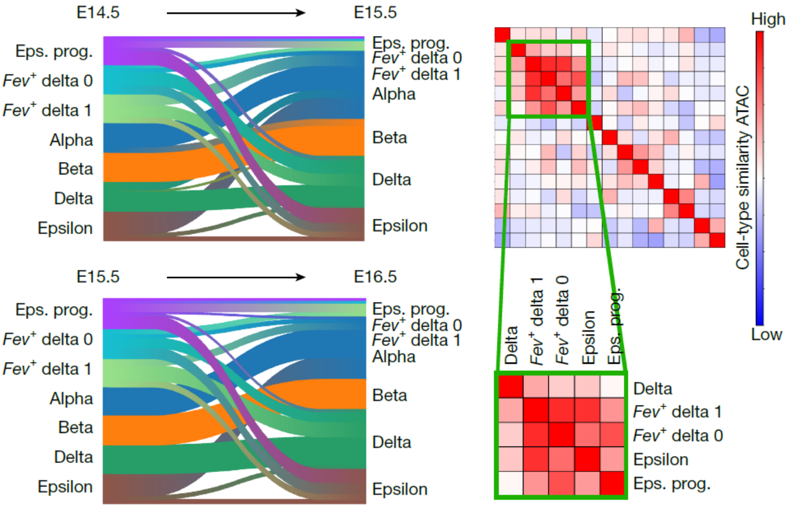
- Cost function based on Euclidean distance in joint embedding performed poorly
- Authors instead used a geodesic cost function

Lineage Segregation of Delta and Epsilon Cells



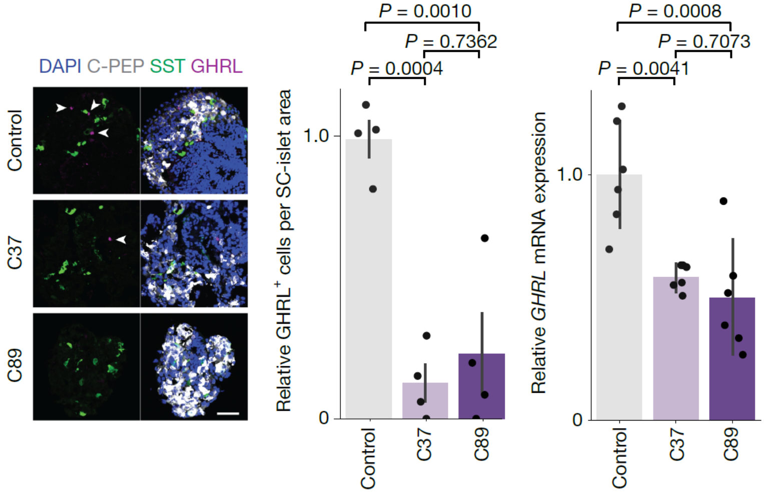
Hypothesis: Delta and epsilon cells follow a similar trajectory.

Corroboration of the Hypothesis



- Eps.prog.: ancestors of epsilon cells
- Fev+ delta: ancestors of delta cells

NEUROD2 as a Regulator of Epsilon Cells



- C37, C89: NEUROD2-KO
- GHRL: marker of epsilon cells

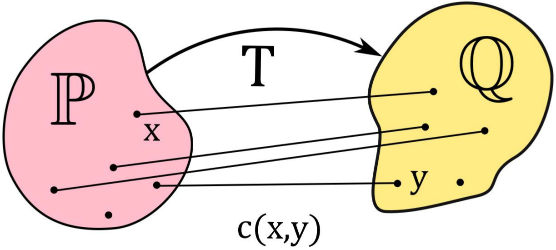
moscot.space.mapping

Mathematical Formulation

$$\arg\min_{P \in U(a,b)} \alpha \sum_{ijkl} L\big(C^X_{ij}, C^Y_{kl}\big)\, P_{ik} P_{jl} + (1-\alpha)\,\langle C, P\rangle - \epsilon H(P)$$

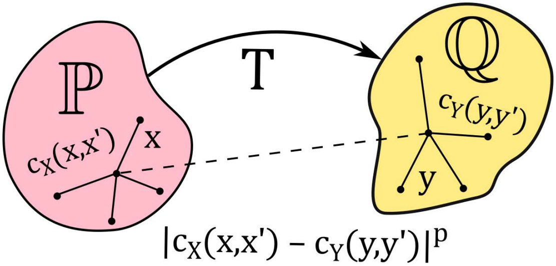
- $U(a,b) := \left\{P \in \mathbb{R}_+^{N\times M} \;\middle|\; P\mathbf{1}_M = a,\; P^\top \mathbf{1}_N = b\right\}$
- $C^X \in \mathbb{R}^{N\times N}$: intra-domain cost (molecular profiles)
- $C^Y \in \mathbb{R}^{M\times M}$: intra-domain cost (spatial distances)
- $C \in \mathbb{R}^{N\times M}$: cross-domain cost (common features)
- α : Balancing parameter between W and GW term
- Can also be extended to unbalanced OT by relaxing marginal constraints with KL divergence

Intuition of W-OT and GW-OT



W-OT

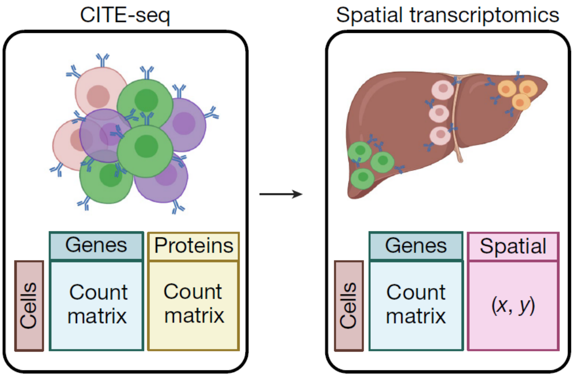
Define Cross-domain Cost

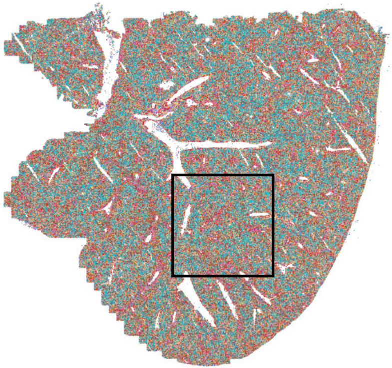


GW-OT

Define Intra-domain Costs

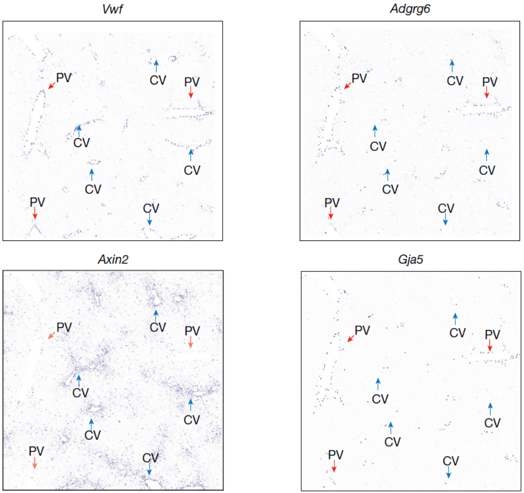
Cross-modal Annotation Transfer for Mouse Liver





Cell-type annotations mapped from single cell reference

Identification of Central and Portal Veins (CVs and PVs)



- **Shared marker**: Vwf (CV + PV, available in ST data)
- **CV marker**: Axin2 (available in ST data)
- **PV markers**: Adgrg6, Gja5 (inferred from CITE-seq reference)

moscot.space.alignment

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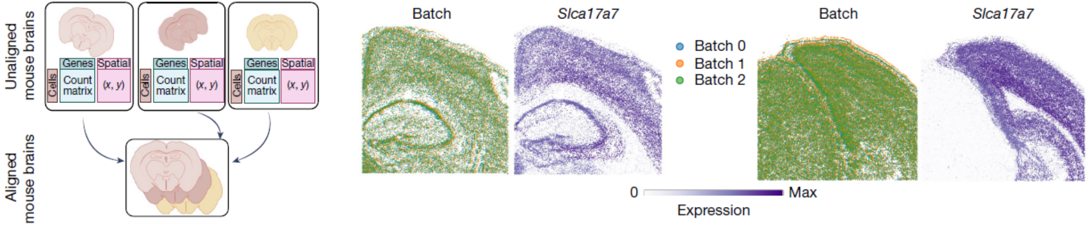
- $C^Y \in \mathbb{R}^{M \times M}$: intra-domain cost (spatial distances)
- $C \in \mathbb{R}^{N \times M}$: cross-domain cost (molecular features)

Aligning Spatial Mouse Brain Sections



- Let Z_1, Z_2 represent the spatial coordinates of adjacent slides
- Obtain the transformed coordinates of slide 2 by

$$\tilde{Z}_2 = P^T Z_1$$

moscot.spatialtemporal

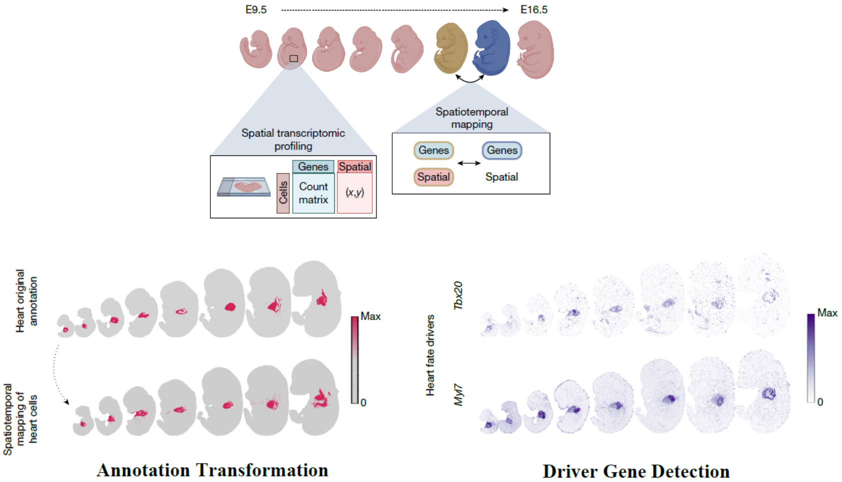
Mathematical Formulation

$$\arg\min_{P \in U(a,b)} \alpha \sum_{ijkl} L\left(C^X_{ij}, C^Y_{kl}\right) P_{ik} P_{jl} + (1-\alpha)\langle C, P\rangle - \epsilon H(P)$$

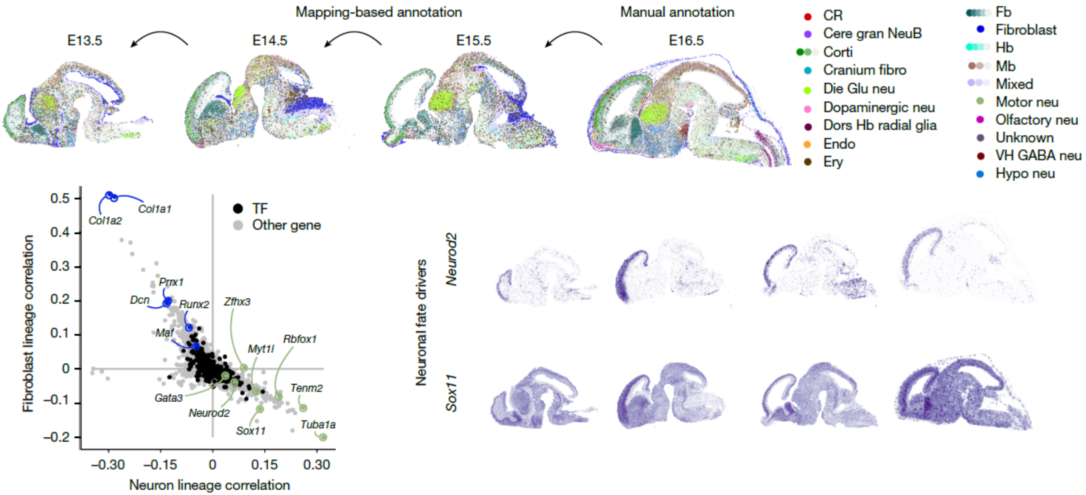
- The objective function is exactly the same as moscot.space.alignment
- moscot.space.alignment: aligns spatial slides across space

  moscot.spatialtemporal: aligns spatial slides across time

Charting spatiotemporal mouse development



Investigating Developmental Trajectories in the Brain



Model Highlights

- Integrates multimodal molecular profiles
- Scales to atlas-sized datasets
- Offers various downstream applications
  - Ancestor/descendant probability inference
  - Driver/target gene detection
  - Cell-type annotation
  - Multimodal imputation
  - Spatial alignment
  - and more

Limitations

- Choice of cost function/embedding is not always robust —requires manual selection
- Hyperparameters (e.g., unbalanced parameters τ_a, τ_b sometimes require manual tuning)
- Discrete OT is in general not applicable to out-of-sample data points (solution: neural OT)
- OT yields only a non-parametric coupling, limiting interpretability of the underlying structure
- Hard to align control vs. perturbed cells due to large distributional gap (solution: neural OT)

Interesting Future Directions

- Investigate unified and biologically interpretable embeddings for calculating cost functions
- Modeling the perturbation effect via neural network on spatially resolved datasets

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Thank You!